ReAdmitRisk: Leakage-Safe 30-Day Readmission Prediction from HL7 FHIR R4 Data

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Abstract

ReAdmitRisk predicts 30-day hospital readmission from HL7 FHIR R4 records and returns each prediction as a FHIR RiskAssessment resource through an API. The system reads FHIR Bulk Data and Bundles, builds its cohort with simplified rules from the Centers for Medicare & Medicaid Services (CMS), and computes 28 features using only information available at discharge. The data were 3,000 synthetic patients with 6,570 index hospitalizations and a 30-day readmission rate of 14.6%. We compared the LACE index, logistic regression, and gradient boosting with monotonic constraints, training on earlier discharges and testing on the latest 1,314 with a 30-day embargo between the two periods. The selected logistic model reached an area under the ROC curve (AUROC) of 0.803 (95% CI 0.758 to 0.839), against 0.752 for LACE (DeLong $p < 0.001$), with a calibration slope of 1.05. A threshold that flags about 10% of discharges caught 34.6% of readmissions, and 52.5% of flagged stays were readmissions. A drift monitor detected a simulated population shift that the Population Stability Index alone did not flag. All results come from synthetic data and say nothing yet about clinical performance.

Keywords: Clinical Prediction Models, Hospital Readmission, HL7 FHIR, Machine Learning Operations, Dataset Shift, Health Informatics.

1 Introduction

Thirty-day readmission is a common target for clinical risk models. A care team can still act after discharge, for example with a follow-up call or an early clinic visit. In the United States, the Hospital Readmissions Reduction Program also lowers Medicare payments to hospitals with more readmissions than expected (Zuckerman et al., 2016).

Most published readmission models discriminate poorly on real data (Kansagara et al., 2011). The LACE index, a four-item score built from length of stay, acuity, comorbidity, and emergency visits, remains a common baseline (van Walraven et al., 2010). A model that scores well on a retrospective split can still fail in use. It may need inputs that hospitals cannot supply in a standard format, or it may learn from information recorded after the discharge it scores (Kaufman et al., 2012). Its calibration can also drift as the patient population changes (Finlayson et al., 2021).

HL7 FHIR gives a standard route to the data. SMART on FHIR defined an apps platform for electronic health records (Mandel et al., 2016), and the FHIR Bulk Data Access API, which certified US health IT must support under the 21st Century Cures Act rule, exports population-level data as NDJSON files (Mandl et al., 2020). Synthea generates synthetic patients in the same format (Walonoski et al., 2018).

This paper describes ReAdmitRisk, an open-source system built around these problems for 30-day all-cause readmission. It reads FHIR R4 records and returns each prediction as a FHIR RiskAssessment resource. We compare two machine-learning models with the LACE index on a temporal split of synthetic data, and we report the data-handling problems found while building the system.

2 Data

2.1 Data Source

The system accepts FHIR R4 as Bulk Data NDJSON files or as Bundles. It uses five resource types: Patient, Encounter, Condition, Observation, and MedicationRequest. Medication codes held in separate Medication resources are resolved, and blood-pressure panels are split into systolic and diastolic values.

Synthea produces realistic FHIR records (Walonoski et al., 2018). In our runs its 30-day readmission rate was only 4 to 6%, and readmissions did not depend on clinical risk factors. We therefore wrote a generator that produces Synthea-shaped records with readmissions drawn from a logistic model. Its risk factors include age, comorbidity, heart failure, prior admissions,

emergency visits, length of stay, polypharmacy, and social needs, and an unobserved frailty term limits how well any model can do. The main dataset has 3,000 patients with records from 2016 through 2025. The parser was tested against trimmed Synthea bundles, and the whole pipeline was run on a 250-patient Synthea bulk export.

2.2 Cohort Construction

Index stays are acute inpatient encounters. Skilled nursing, rehabilitation, detoxification, hospice, and psychiatric stays are excluded both as index stays and as readmissions. Encounters that start within 24 hours of the previous one ending are merged into one hospitalization; Synthea, for example, records one ICU stay as three encounters (admission, ICU transfer, and ward). We excluded patients under 18, deaths during the stay, and discharges within 30 days of the data extraction date. The outcome is any acute hospitalization starting within 30 days of discharge. The CMS hospital-wide measure removes planned readmissions with a dedicated algorithm (Horwitz et al., 2014); that algorithm was not implemented here.

2.3 Cohort Overview

Of 8,166 inpatient encounters, 7,570 were acute. Merging 734 transfer encounters into the stays they continued left 6,836 hospitalizations. Removing 254 stays that ended in death and 12 without full follow-up left 6,570 index hospitalizations from 2,203 patients, of which 959 (14.6%) were followed by a readmission. The training period held 5,205 stays (readmission rate 14.8%) and the test period 1,314 (14.1%); the embargo removed 51 stays.

Table 1 compares readmitted and other patients using one randomly chosen stay per patient. Counting every stay as independent made sex and race appear associated with readmission ($p = 0.0003$ and $p = 0.0007$), although the generator gives neither any effect. With one stay per patient the unadjusted p-values were 0.67 and 0.087.

3 Methods

3.1 Feature Engineering

All 28 features are computed as of the discharge time. Utilization counts end at the admission, and the emergency visit that led to the admission does not count as a prior visit. A condition counts only once it was recorded, which keeps a diagnosis documented later with an earlier onset date out of training. Medication counts use prescriptions written in the prior 12 months and ignore the request status, which describes the record at export time. These rules target leakage, where training data carry information that would not exist at prediction time (Kaufman et al., 2012). A property test computes features on full records and on the same records cut off at discharge, and requires the two to match.

Comorbidity is summarized with the Charlson Comorbidity Index (Charlson et al., 1987), mapped from SNOMED CT and ICD-10 display text. The rules were checked against conditions Synthea emits; suspected cancer and carcinoma in situ, for instance, do not count as malignancy. The LACE score (van Walraven et al., 2010) was computed from the same inputs.

3.2 Models

The candidates were the LACE score mapped to a probability, L2-regularized logistic regression on all features, and histogram gradient boosting. The boosting model used monotonic constraints, so predicted risk cannot fall when prior admissions, emergency visits, comorbidity, or length of stay rise. The selected model was calibrated with Platt scaling fitted on held-out folds (Niculescu-Mizil and Caruana, 2005).

3.3 Validation Design

The latest 20% of discharges formed the test set. Training stays discharged in the 30 days before the test period were dropped, because their labels depend on test-period events. Hyperparameters were tuned with 5-fold cross-validation grouped by patient, and the model with the best cross-validated average precision was selected. The high-risk threshold was set to flag 10% of training-period discharges and then applied unchanged to the test period.

3.4 Statistical Analysis

Confidence intervals for the AUROC, average precision, and Brier score come from 1,000 bootstrap samples of patients rather than stays, since stays from the same patient are correlated (Field and Welsh, 2007). AUROCs were compared with DeLong's test for paired ROC curves (DeLong et al., 1988). Calibration was summarized by the slope and intercept of a logistic recalibration and by the expected calibration error (ECE) over ten bins (Van Calster et al., 2019); observed rates per decile carry Wilson score intervals (Wilson, 1927). Cohort characteristics were compared with Mann-Whitney U, chi-square, or Fisher's exact tests, with Holm correction across variables. A quality gate required a test AUROC of at least 0.65 and an improvement over LACE before a model could be saved.

Table 1: Cohort characteristics, one randomly chosen stay per patient. Values are median [interquartile range] or n (%). P-values are Holm-adjusted across variables. ED, emergency department; CKD, chronic kidney disease; eGFR, estimated glomerular filtration rate.

Characteristic	Readmitted ($n = 150$)	Not readmitted ($n = 2{,}053$)	p
Age, years	84.0 [76.1–94.2]	76.2 [62.3–86.2]	< 0.001
Male sex	70 (46.7)	1,002 (48.8)	1.00
Length of stay, days	3.9 [2.6–5.8]	3.0 [2.1–4.5]	< 0.001
Admitted through the ED	109 (72.7)	1,281 (62.4)	0.11
Admissions in the prior 12 months	0 [0–2]	0 [0–0]	< 0.001
ED visits in the prior 6 months	0 [0–1]	0 [0–0]	< 0.001
Charlson Comorbidity Index	2 [1–4]	1 [0–2]	< 0.001
Congestive heart failure	46 (30.7)	228 (11.1)	< 0.001
Moderate or severe CKD	54 (36.0)	228 (11.1)	< 0.001
Polypharmacy	82 (54.7)	560 (27.3)	< 0.001
Latest eGFR, mL/min/1.73 m^2	72.5 [46.8–85.6]	83.2 [72.6–91.9]	< 0.001
LACE index	9 [6–11]	6 [4–7]	< 0.001
Race (six groups)			0.44

3.5 Deployment and Monitoring

Figure 1 shows the system. A FastAPI service computes features from a submitted FHIR Bundle with the same code used in training and returns a RiskAssessment resource. It also exposes Amazon SageMaker's hosting endpoints, so one container image runs on Kubernetes or SageMaker. Runs and models are tracked in MLflow. A care-transition assistant writes briefs with retrieval-augmented generation (Lewis et al., 2020); each brief must cite the FHIR resources it relies on, and citations that match no supplied resource are removed.

Drift is measured on every feature and on predicted risk. A feature is flagged when its Population Stability Index (PSI) reaches 0.2 or its standardized mean difference (SMD), a scale-free measure of the difference between two groups (Austin, 2009), reaches 0.2 in absolute value.

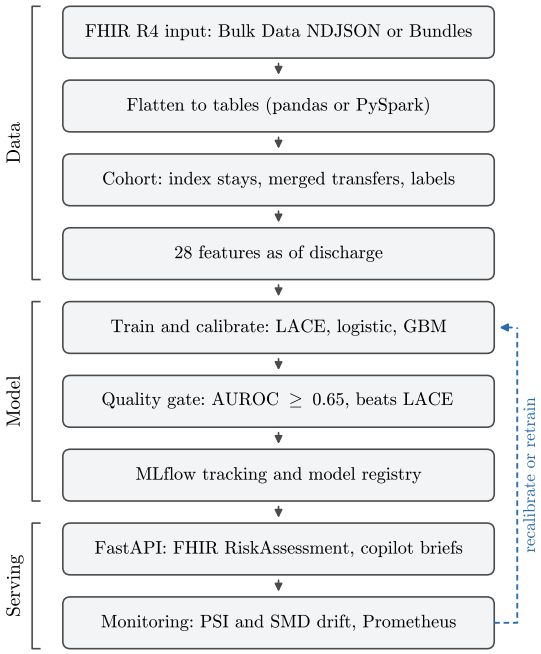


Figure 1: System overview, from FHIR input to monitoring. The dashed path marks the response when monitoring detects a shift.

4 Results

4.1 Discrimination and Calibration

Logistic regression had the best cross-validated average precision and outperformed LACE on the test period (Table 2). Its AUROC had a 95% confidence interval of 0.758 to 0.839, and its gain over LACE was 0.051 (95% CI 0.024 to 0.078; DeLong $p = 0.0002$). Gradient boosting scored 0.005 higher on the test set, a difference that was not significant ($p = 0.36$) and that cross-validation did not favor. Brier scores were 0.099 for logistic regression, 0.098 for gradient boosting, and 0.107 for LACE.

Predicted risk averaged 15.4% in the test period against an observed rate of 14.1%, giving a calibration intercept of -0.13 and a slope of 1.05. Nine of the ten decile intervals contain the predicted rate; the exception is a decile with no readmissions (Figure 2). Permutation importance ranked emergency visits in the prior six months first, followed by prior admissions and length of stay.

4.2 Operating Point

At the threshold of 0.361, the model flagged 122 test stays (9.3%). Those stays included 64 of the 185 readmissions, a sensitivity of 34.6%. Of the flagged stays, 52.5% were readmissions, 3.7 times the base rate, and specificity was 94.9%.

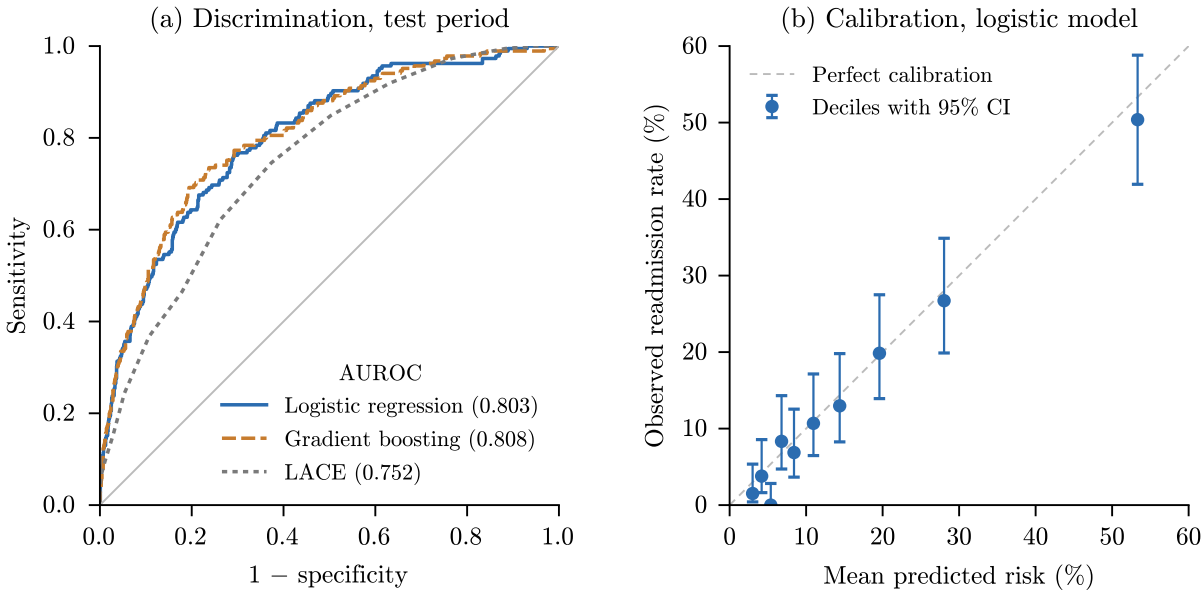


Figure 2: Test-period performance: (a) ROC curves for the three models, with AUROC in the legend; (b) observed readmission rate against mean predicted risk by decile for the logistic model, with 95% Wilson intervals.

Table 2: Cross-validated (CV) and held-out performance on 1,314 test stays. AP is average precision; slope is the calibration slope.

Model	CV AP	AUROC	AP	Slope	ECE
LACE	0.386	0.752	0.340	0.99	0.019
Logistic (selected)	0.478	0.803	0.446	1.05	0.017
Gradient boosting	0.462	0.808	0.454	1.07	0.021

Table 3: The two calibration methods on the same data and test period.

Method	AUROC	Brier	ECE	Slope
Platt (selected)	0.803	0.0988	0.017	1.05
Isotonic	0.801	0.0989	0.020	0.97

4.3 Calibration Method

Isotonic regression in place of Platt scaling made no meaningful difference, so Platt scaling was kept (Table 3). Isotonic regression had a slope closer to 1 but a higher expected calibration error, and the other measures were nearly equal.

4.4 Subgroup Audit

AUROC was similar for women and men (0.807 and 0.797) and across ethnicity (gap 0.004). The larger gaps across race (0.202) and age band (0.141) came from small groups: Asian patients (84 stays, about 11 readmissions; AUROC 0.611) and ages 65 to 74 (143 stays, about 6 readmissions; AUROC 0.664). The generator has no race effect, so the race gap most likely reflects sampling variation. For patients recorded as Other race, observed readmissions were 0.54 times the predicted number. These comparisons count stays rather than patients, so their p-values ($p = 0.011$ for equal sensitivity across race) share the clustering problem described in Section 2.3.

4.5 Interoperability and Serving

The pipeline ran on a 250-patient Synthea v3.3.0 bulk export. Flattening its 284 patient records and 134,121 observations took 8.9 seconds on one CPU core. All medication names were resolved, and race was parsed for 98.6% of patients. Cohort rules removed 80 non-acute stays and merged 12 transfer chains, leaving 198 index stays with 12 readmissions (6.1%), too few to train on. On a laptop, the API answered feature-based requests in about 90 ms and FHIR Bundle requests in about 120 ms.

4.6 Drift Monitoring

The monitor flagged 12 features in a simulated population of 1,000 older, sicker patients with 3,395 index stays. No feature crossed the PSI threshold, so every flag came from the SMD criterion (Figure 3). Predicted risk rose from 14.9% to 22.2% on average (PSI 0.152, SMD 0.36), and emergency visits per stay nearly doubled, from 0.54 to 1.05 (PSI 0.135). PSI was 0.000 for binary features even where prevalence doubled, as for cancer (7.2% to 15.4%). The model still ranked patients well on the shifted population (AUROC 0.823), but the observed readmission rate of 25.0% exceeded the mean prediction of

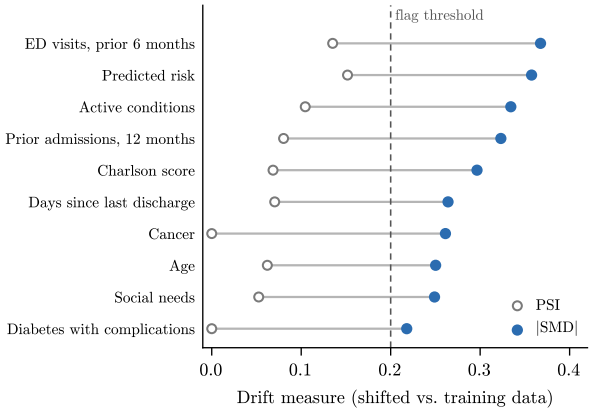


Figure 3: PSI and absolute SMD for the ten features that shifted most in the simulated population. The dashed line marks the 0.2 flag threshold used for both measures.

22.2%.

5 Discussion

On synthetic data, regularized logistic regression beat the LACE index by 0.051 AUROC and stayed well calibrated on later discharges. Gradient boosting added nothing measurable. That fits the generator, whose effects are additive on the log-odds scale apart from one interaction between heart failure and age. The features ranked most important were among the generator's strongest risk factors, so the pipeline recovered the signal it was given, which is the most a synthetic evaluation can show.

5.1 Problems Found During Development

Several data-handling issues would have biased the results if left alone. Without merging, the 734 transfer encounters would have become separate index stays, and a transfer that began after the previous leg ended would have been labeled a readmission. Two leakage paths were specific to FHIR: onset dates can be recorded after the fact, and MedicationRequest status describes the record at export time. Either would raise training performance with information unavailable at discharge. Text-based comorbidity rules needed tests against real condition names; one early rule matched the phrase "Hearing aids user" as AIDS, which would have added 6 points to the Charlson score, and a test caught it. Repeated stays made unrelated attributes look significant in the cohort comparison until each patient contributed a single stay. The subgroup tests in Section 4.4 count stays as well and need the same patient-level treatment.

5.2 Monitoring

PSI alone would not have raised a drift alert for a population whose mean predicted risk rose by half. Its quantile bins collapse on counts dominated by zeros and on binary features. Pairing it with the standardized mean difference flagged the 12 features that shifted most. The performance check did not recommend retraining, because discrimination held at 0.823. Predictions still ran 2.8 points below the observed rate, so a care team would see more readmissions than the model flagged, the kind of silent degradation that clinicians are asked to watch for (Finlayson et al., 2021). Recalibrating on recent data would be the proportionate response.

5.3 Limitations

All data were synthetic, so the results describe the pipeline; clinical performance remains untested, and published models on real data have mostly shown weaker discrimination (Kansagara et al., 2011). The CMS planned-readmission algorithm was not applied, and readmissions to hospitals outside the data source cannot be seen. Race and ethnicity were generated independently of risk, so the subgroup results test the audit itself and say nothing about real disparities. Readmission measures use of care, which also reflects access to care; a label of this kind produced racial bias in a widely used algorithm trained on health care costs (Obermeyer et al., 2019). Before any clinical use, a deployment would need local validation with subgroup confidence intervals and a clinical review.

5.4 Future Work

Validation on real records is the next step. MIMIC-IV is available in FHIR form (Bennett et al., 2023) and requires credentialed access through PhysioNet. Other directions are a planned-readmission filter, subgroup intervals from the patient-level bootstrap, and a recalibration job triggered by the drift monitor.

6 Conclusion

ReAdmitRisk turns FHIR R4 records into calibrated readmission predictions returned as FHIR resources. On synthetic data its model beat the LACE index, with an AUROC of 0.803 against 0.752. The largest risks to validity came from data handling: transfer encounters, record timestamps, repeated stays, and the choice of drift metric. Each was caught by a test or a check, and the fixes are part of the pipeline.

Acknowledgments

Test fixtures were generated with Synthea, an open-source synthetic patient generator (Walonoski et al., 2018).

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